

Tut - 3

Q.14 Define order and degree of a differential equation.

Order: The order of a differential equation is the ^{degree} power of the highest derivative occurring in it.

Degree: The degree is the power of highest order derivative when the equation is expressed as polynomial.

Q.21 Write the standard form of linear differential eq.

$$\frac{dy}{dx} + p(x)y = Q(x)$$

Q.31 What is integrating factor? Write the I.F for

$$\frac{dy}{dx} + p(x)y = Q(x)$$

$$I.F = e^{\int p \cdot dx}$$

Q.41 Define an exact differential equation and state the condition of existence.

A differential eq $M(x, y) dx + N(x, y) dy = 0$

Condition for exactness $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Q.51 Write the standard form of Bernoulli's diff. eqn.

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Part - B

B1: Solve $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$

→ This eqn is in form of $M(x, y)dx + N(x, y)dy = 0$

$$M = 2xy + y^2, N = x^2 + 2xy$$

$$\frac{\partial M}{\partial y} = 2x + 2y, \frac{\partial N}{\partial x} = 2x + 2y$$

Since, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Integrate: $\int M dx$ (keeping y constant) + $\int (N \text{ without } x) dy = C$

$$\int (2x + y^2) dx + \int 0 = C$$

$$\int 2x dx + \int y^2 dx = C$$

$$2x \cdot \frac{x^2}{2} + y^2 x = C$$

$$x(x^2 + y^2) = C$$

$$x(x^2 + y^2) = C$$

B2: Solve $\frac{dy}{dx} + y \tan x = \sin x$

→ eqn is comparable with standard equation of linear differential eqn:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $p = \tan x$ & $Q = \sin x$

$$I.F = e^{\int \tan x \cdot dx}$$

$$= e^{\log |\sec x|}$$

$$= \sec x$$

$$\text{Solution :- } y \times \sec x = \int \sin x \times \sec x \cdot dx$$

$$y \times \sec x = \int \sin x \times \frac{1}{\cos x} \cdot dx$$

$$y \times \sec x = \int \tan x \cdot dx$$

$$y \times \sec x = \log |\sec x| + \log c$$

$$y = \cos [\log |\sec x| + \log c]$$

32. Solve $\frac{dy}{dx} + y = y^2$

$\frac{1}{y^2}$ dividing whole eq. by y^2

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{y} = 1$$

$$\frac{1}{y} = t$$

$$-\frac{1}{y^2} \cdot \frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} + t = 1$$

$$\frac{dt}{dt} - t = 1 \quad \text{homogeneous} \quad \text{particular}$$

Now eqn is in standard form of L.D.E

$$I.F = e^{-\int 1 dt}$$

$$I.F = e^{-t}$$

Solution:-

$$t \times e^{-t} = \int e^{-t} \cdot dt$$

$$t \times e^{-t} = -e^{-t} + C$$

putting $t = \frac{1}{y}$

$$\frac{1}{y} \times e^{-1/y} = -e^{-1/y} + C$$

Part - C

Solve the differential eqn $(x^2y + y^3) dx + (x^3 + 3xy^2) dy = 0$

$\therefore$ equation is in form of $M(dx) + N(dy) = 0$

$$M = x^2y + y^3$$

$$N = x^3 + 3xy^2$$

$$\frac{\partial M}{\partial y} = x^2 + 3y^2$$

doubt

$$\frac{\partial N}{\partial x} = 3x^2 + 3y^2$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$c2: \frac{dy}{dx} + \frac{y}{x} = x^2 y^2$$

→ The eqⁿ is in Bernoulli's form:
divide by y^2 :

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{y} = x^2$$

$$\text{let } \frac{1}{y} = t$$

$$-\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} + \frac{t}{x} = x^2$$

$$\frac{dt}{dx} - \frac{t}{x} = -x^2$$

IIT
PAGE No.
DATE

$$\begin{aligned} \text{I.F.} &\rightarrow e^{\int -\frac{1}{x} dx} \\ &= e^{-\log x} \\ &= e^{\log x^{-1}} \\ &= \frac{1}{x} \end{aligned}$$

Solution:- $\log x = -\int \frac{x^2 x}{x^2} dx$

$$\frac{1}{x} \cdot \frac{1}{x} = -\int 1 dx$$

$$\frac{1}{x} = -\frac{x^2}{2} + C$$

$$1 = -\frac{x^3}{2} + Cx$$

$$\frac{1}{y} = -\frac{x^3}{2} + Cx$$

$$y = \frac{1}{Cx - \frac{x^3}{2}}$$

Q3: Solve $x \frac{dy}{dx} + y = x \cos x$

Soln:- $\div x$

$$\frac{dy}{dx} + \frac{y}{x} = \cos x$$

$$\begin{aligned} \text{I.F.} &\rightarrow e^{\int \frac{1}{x} dx} \\ &= e^{\log x} \\ &= x \end{aligned}$$

Soln:- $y \cdot x = \int x \cdot \cos x \cdot dx$

$$y \cdot x = \int x \cos x - \int \frac{dx}{x} \cdot x \cos x$$

$$y'x = x \int \cos x \cdot dx - \int \frac{dx}{dx} \int \cos x \cdot dx$$

$$y'x = x(\sin x) - \int \sin x \cdot dx$$

$$y'x = x(\sin x) + \frac{\cos x}{1} + C$$

$$y = x \sin x + \frac{\cos x}{1} + \frac{C}{1}$$